Ethanol Blending in Gasoline – Thailand

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U.S. GRAINS & BIOPRODUCTS COUNCIL

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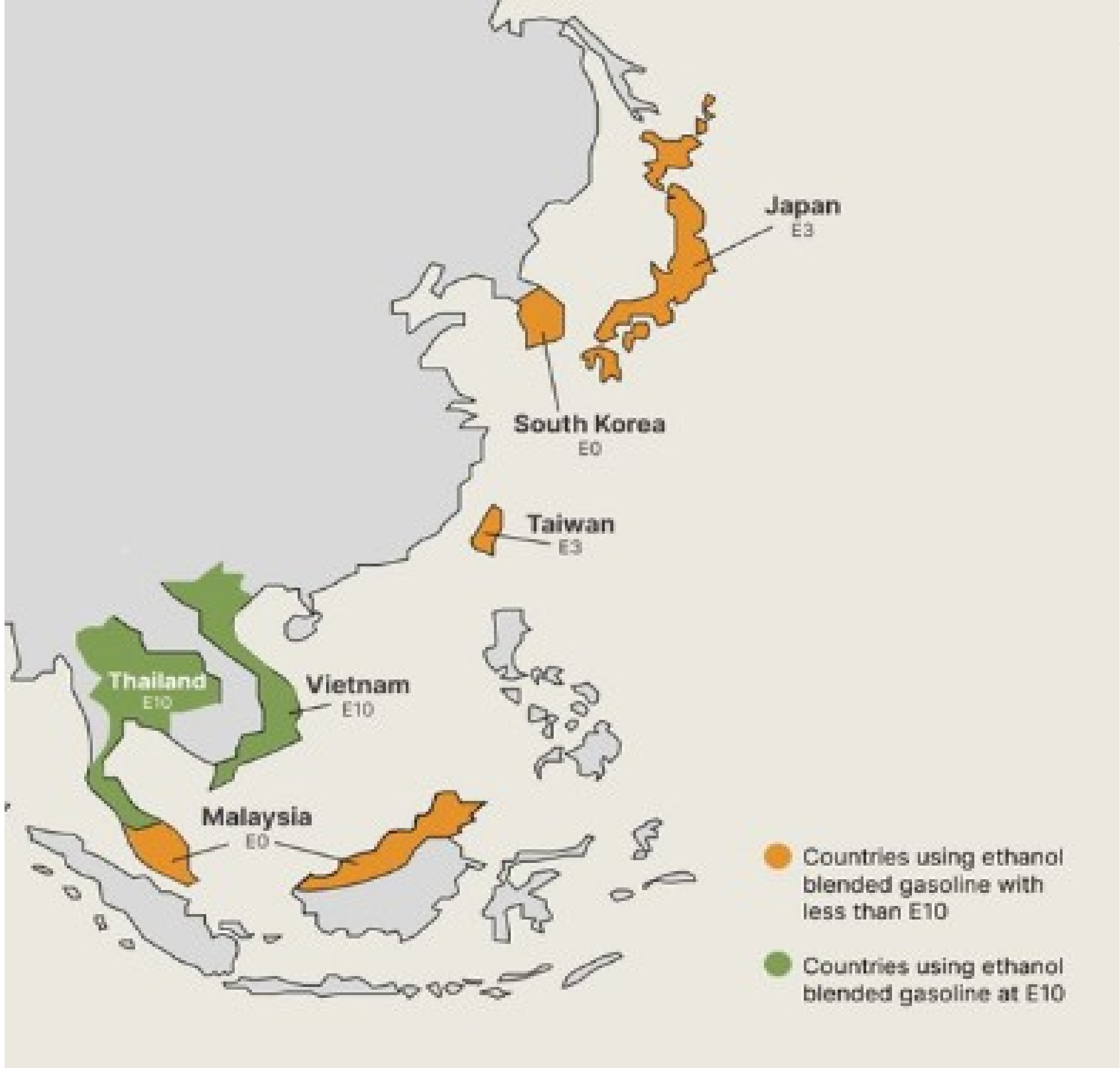
Ethanol Blending in Asia

There are important fuel quality and environmental impact of vehicle emission challenges in these regions.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Asia to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Seven countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

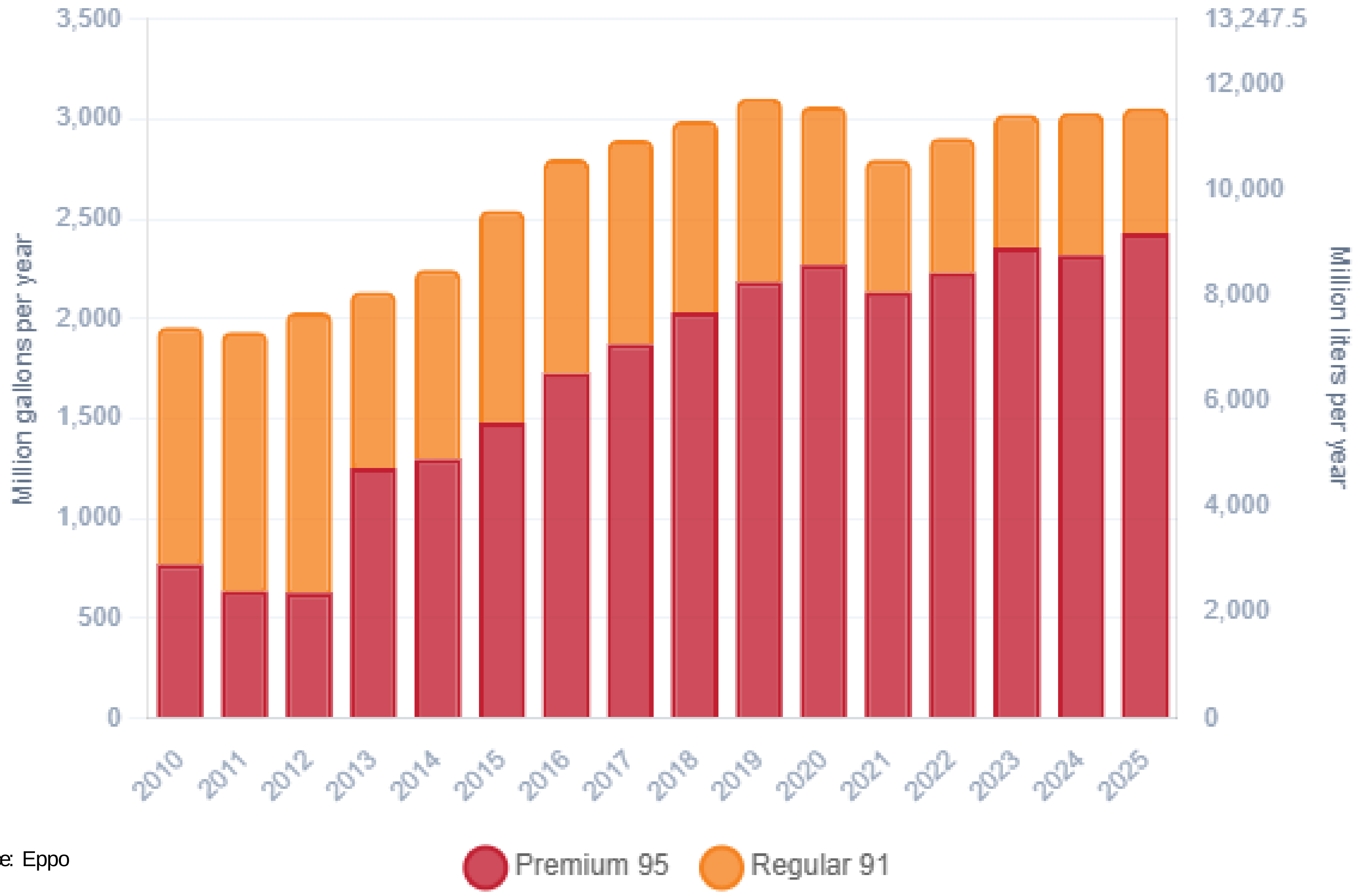
Ethanol Gasoline Blending in Thailand

In 2025, gasoline consumption in Thailand was 11,576 million liters (3,448 million gallons) and growth rate was 3.0%. Gasoline production and net imports were 13,051 and -1,312 million liters (3,448 and -347 million gallons) correspondingly. Thailand offers two main grades: RON 91 (E10) and RON 95 (E10 and E20) with an average octane of 94.2 RON.

Thailand's bioethanol production primarily uses sugarcane and cassava as feedstocks, with approximately 30 facilities producing over 1 billion liters annually. The promotion of bio-based products like ethanol is part of Thailand's bioeconomy policy, supporting both the energy sector and domestic economic growth. In 2025, ethanol consumption supplied by local production in Thailand was 1,290 million liters (341 million gallons) representing 11.0% of gasoline demand.

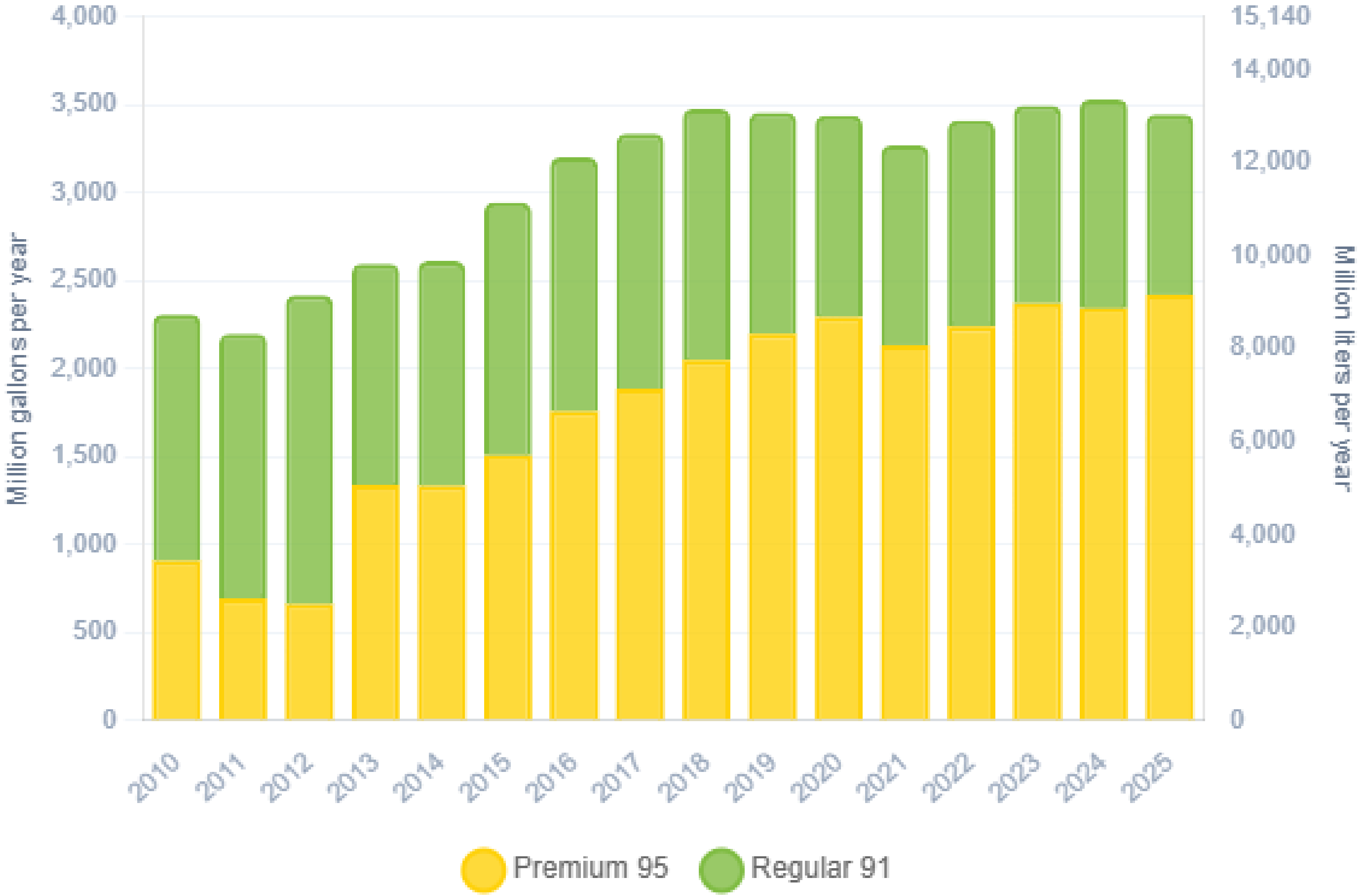
Source: Eppo, USDA, Department of Energy Buisiness

Gasoline Demand in Thailand



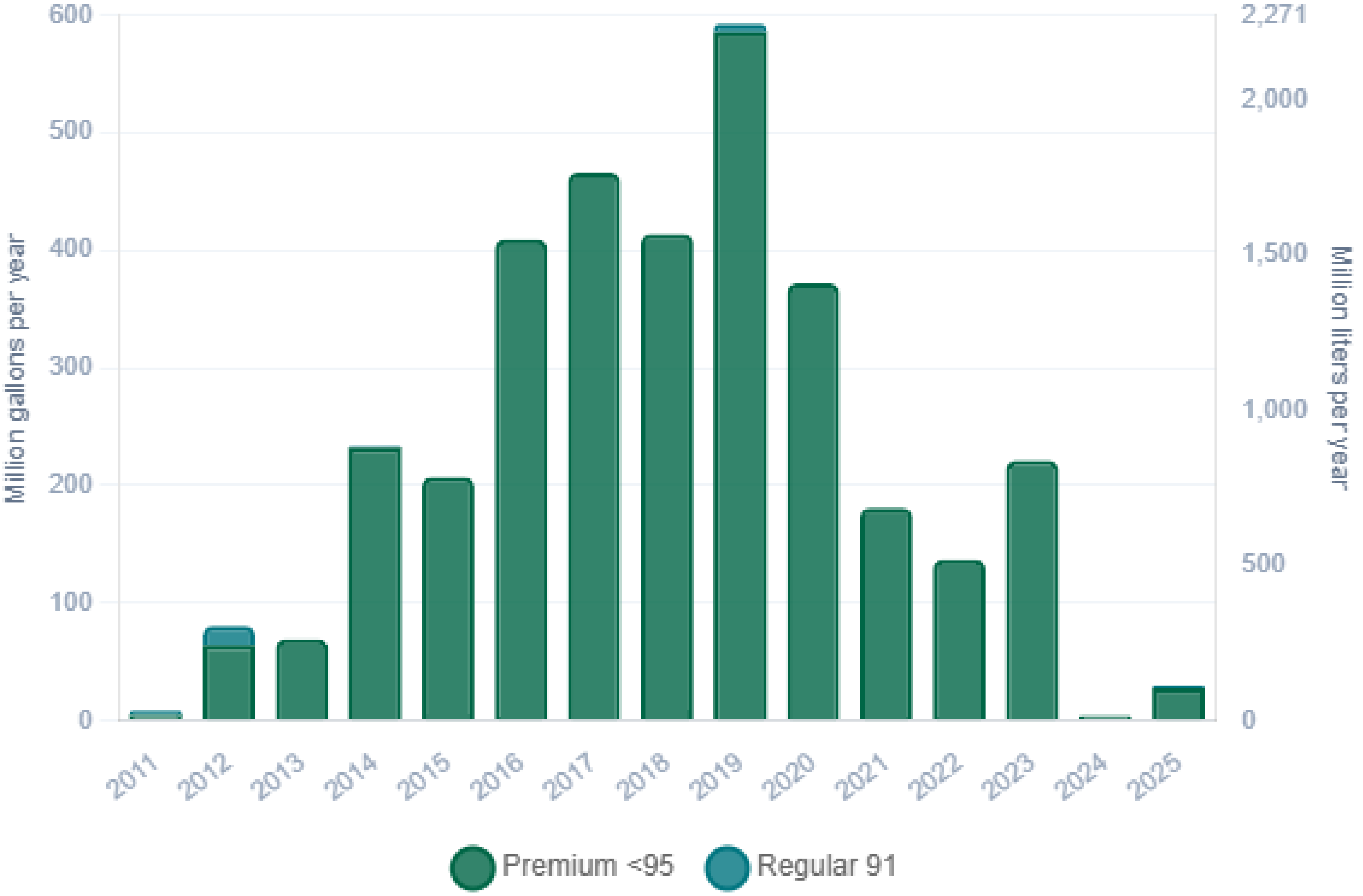
Source: Eppo

Gasoline Production in Thailand



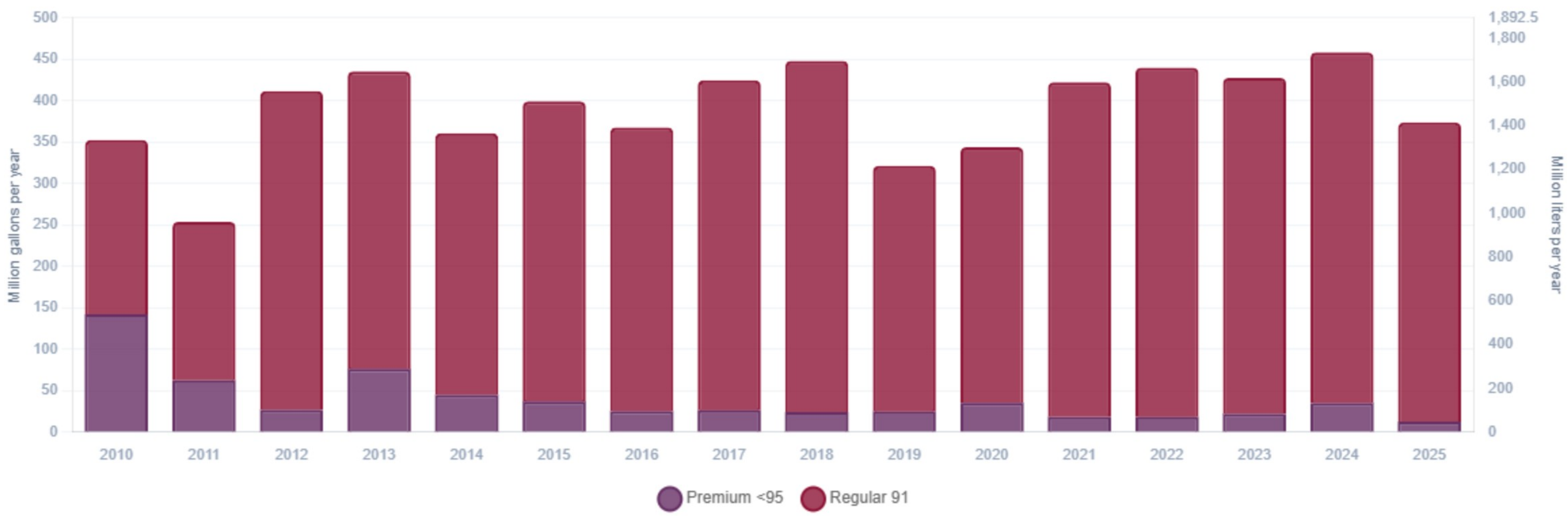
Source: Eppo

Gasoline Imports in Thailand



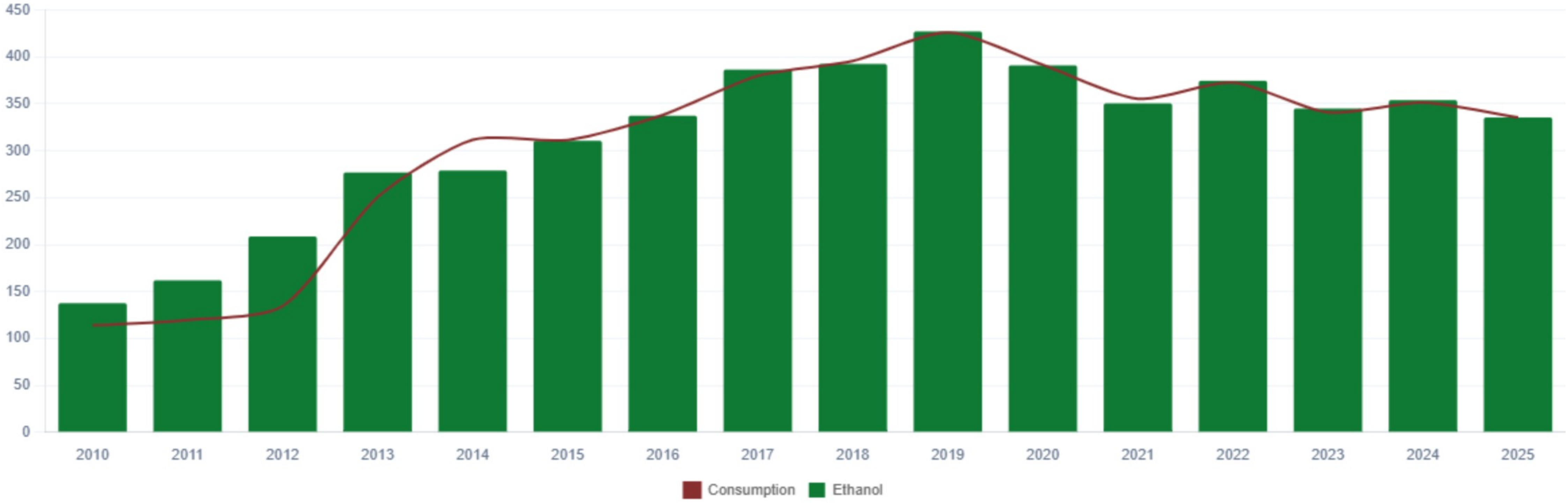
Source: Eppo

Gasoline Exports in Thailand



Source: Eppo

Ethanol Balance in Thailand



Source: Eppo, USDA

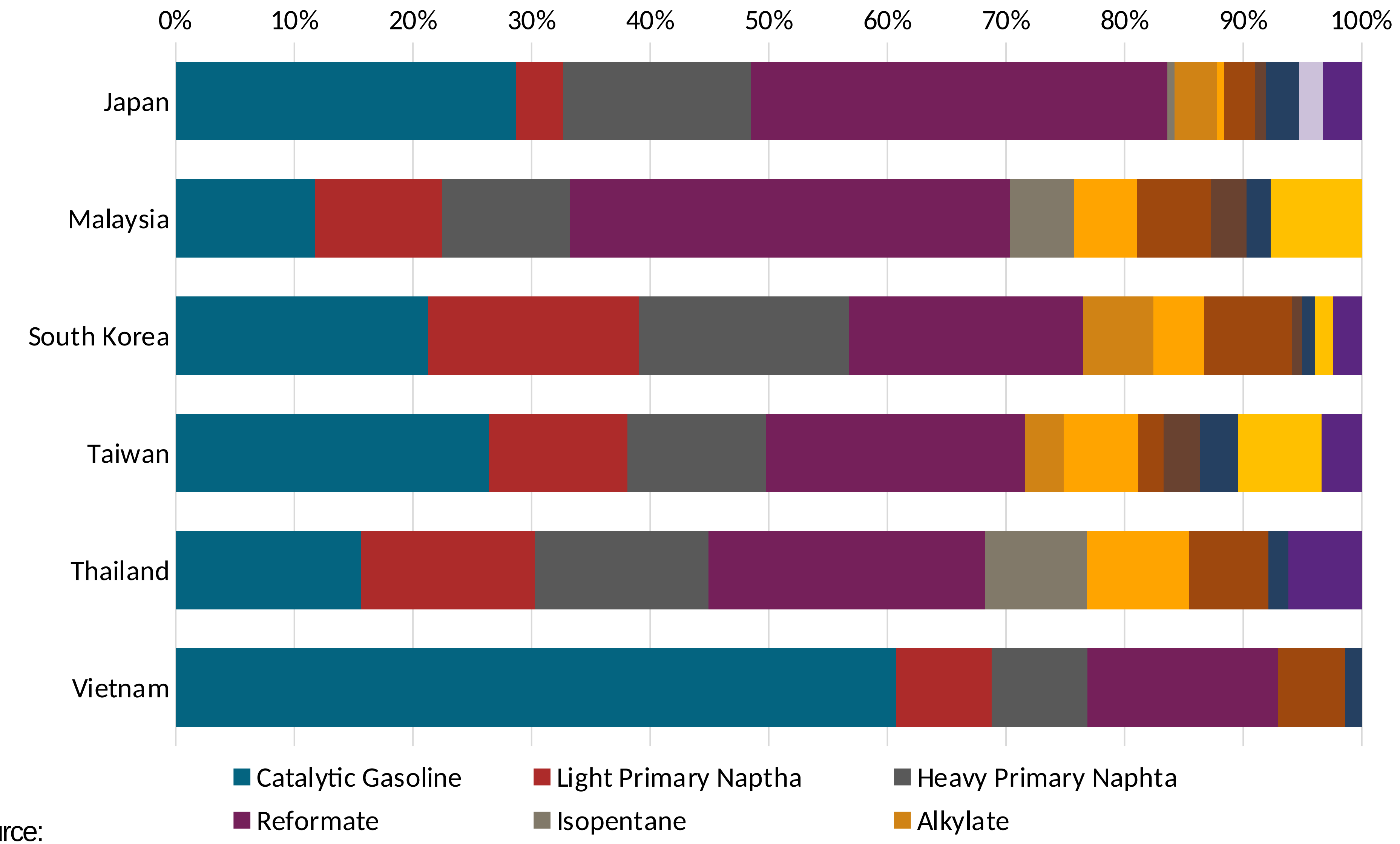
Gasoline Quality in Thailand

Source: Department of Energy Buisiness

Gasoline Component Blending in Asia

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Asia are:

- Catalytic Gasoline
- Light Primary Naptha
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics



Source: Faro90

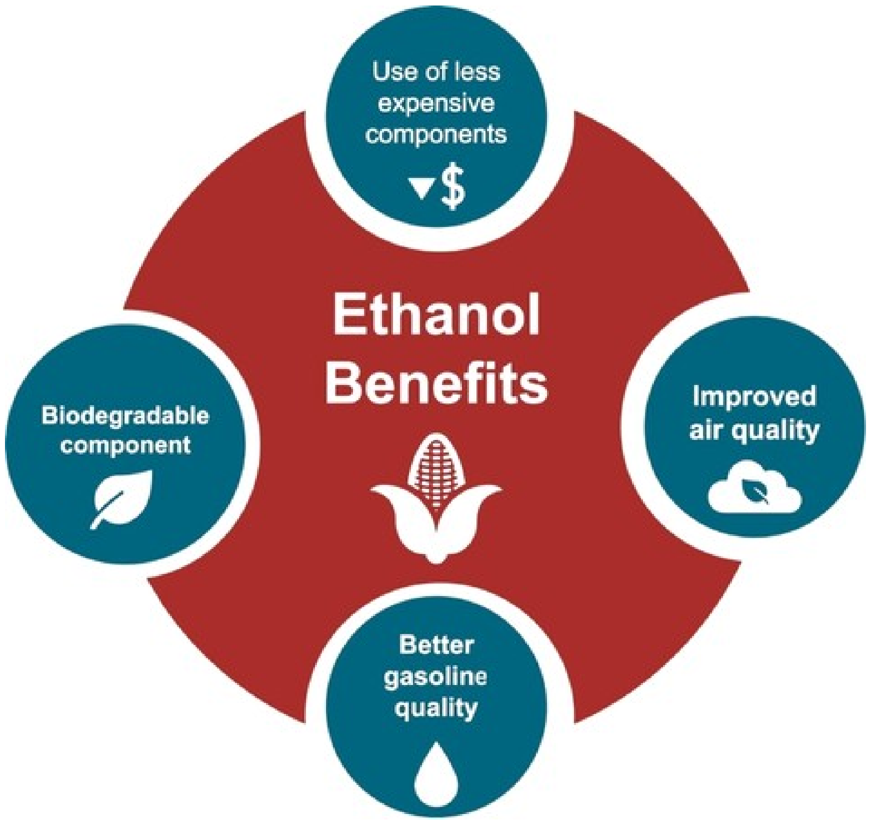
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

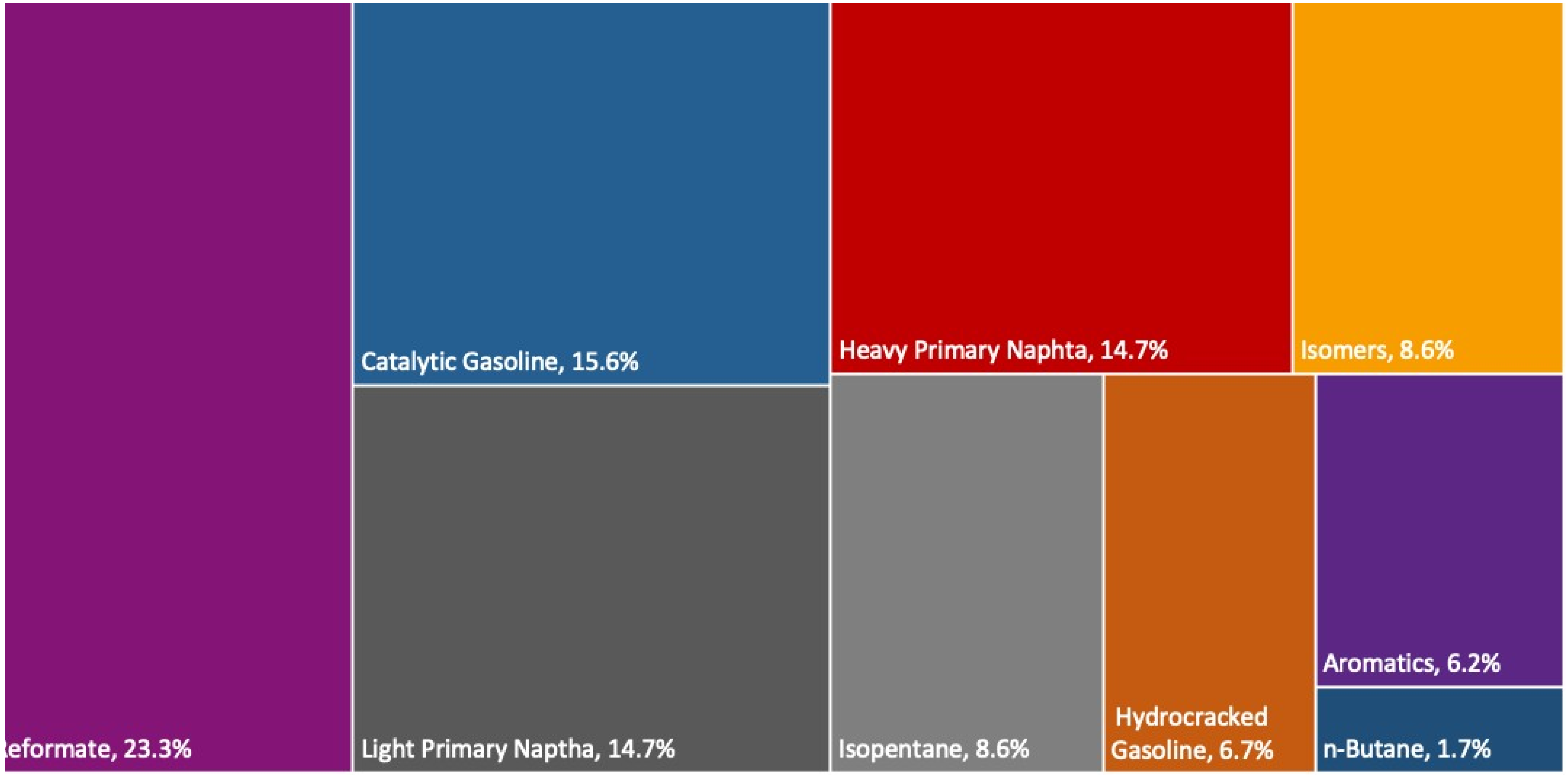
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

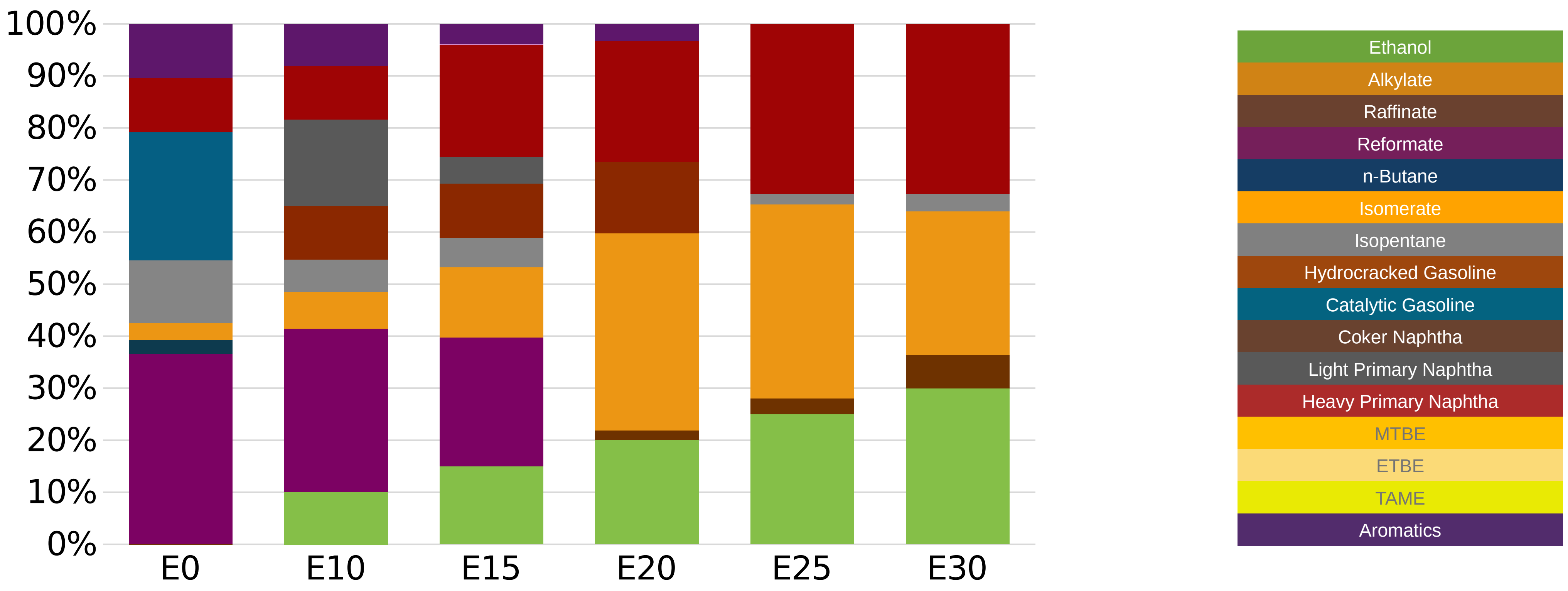


Thailand Available Blending Components



Source:
Faro90

Thailand – Gasoline 91 – Constant Octane

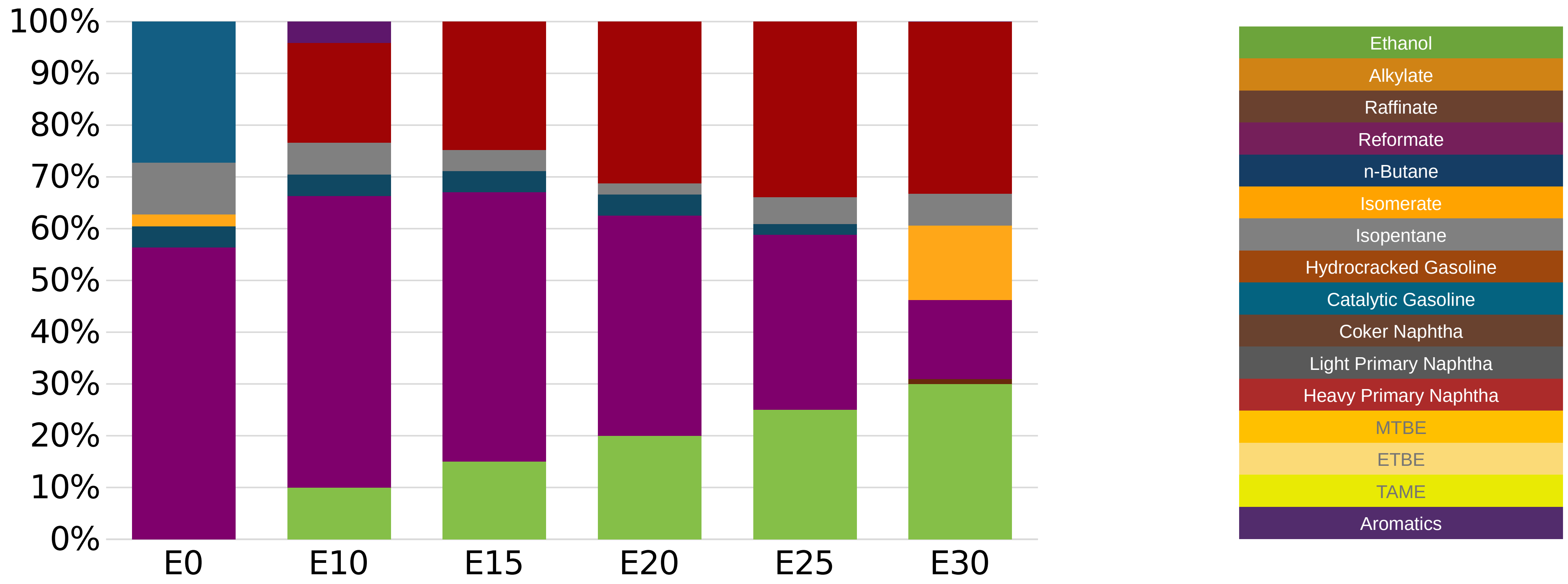


Octane (RON)	91.0	91.0	91.0	91.4	91.6	93.1
Price (US$/gal)	$ 2.043	$ 1.910	$ 1.794	$ 1.734	$ 1.595	$ 1.553

Average CIF 2025 Prices

Source: Faro90

Thailand - Gasoline 95 - Constant Octane

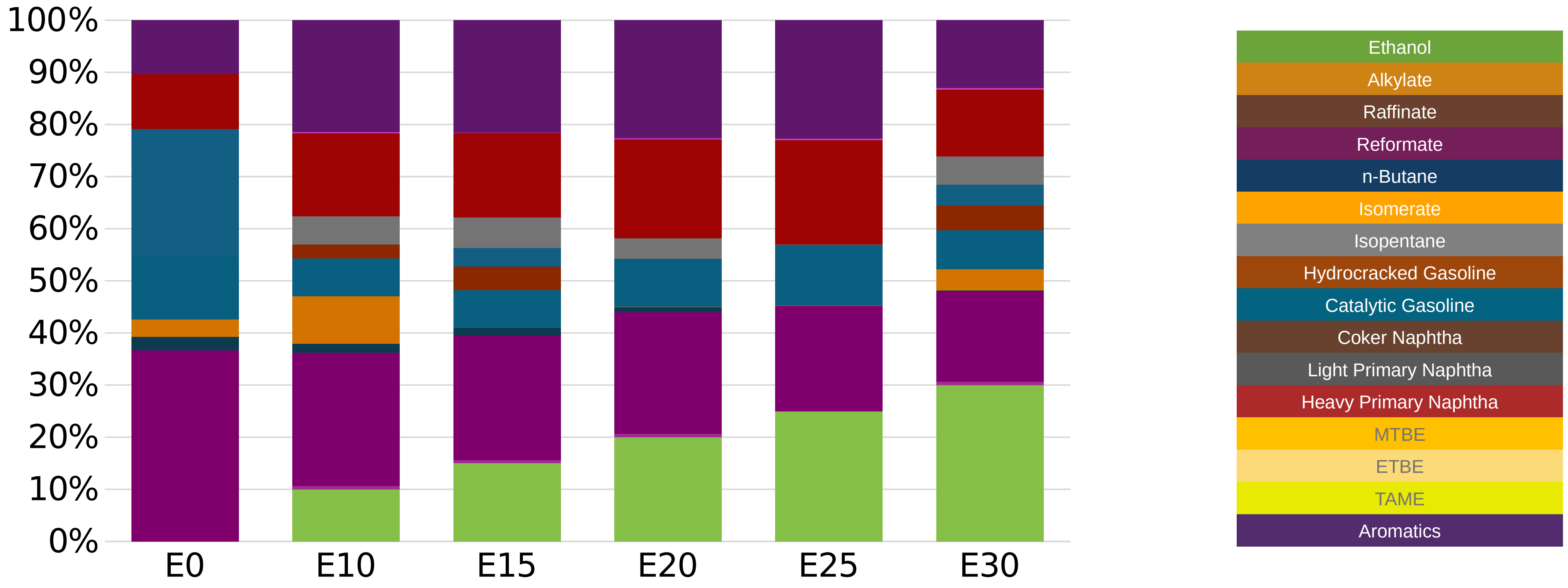


Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (US$/gal)	$ 2.243	$ 2.063	$ 1.966	$ 1.862	$ 1.755	$ 1.660

Average CIF 2025 Prices

Source: Faro90

Thailand – Gasoline 91 – Increased Octane

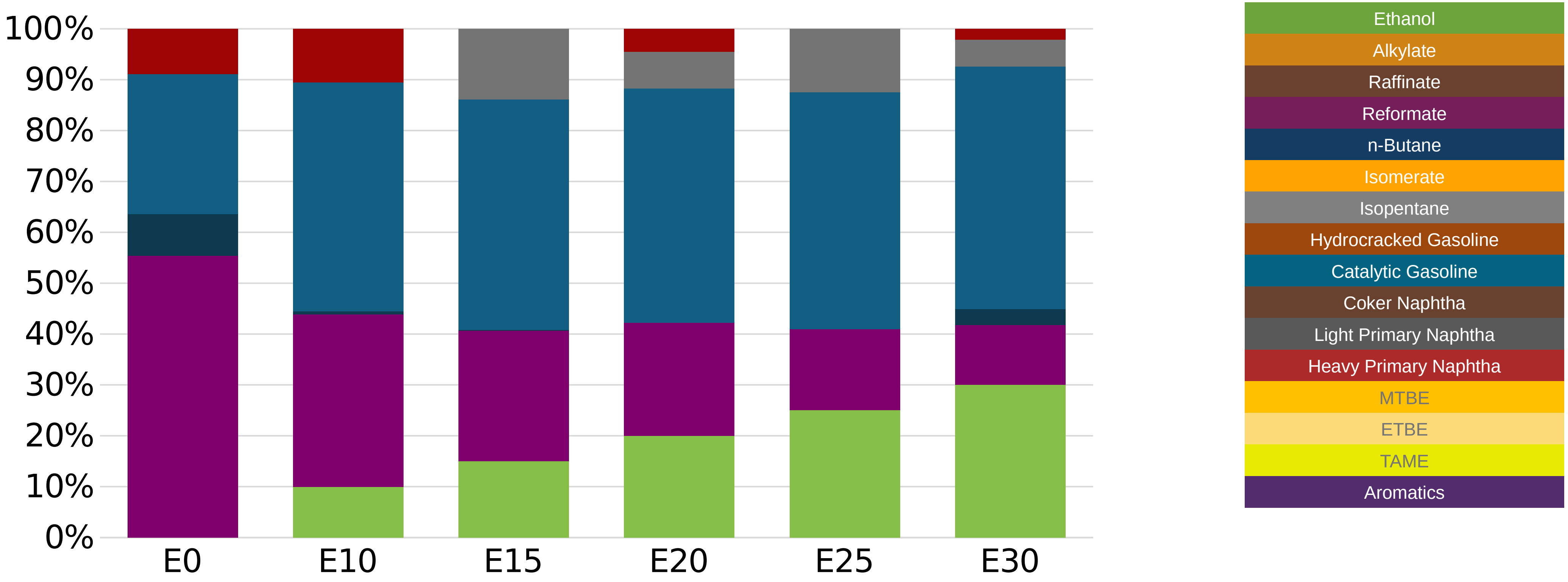


Octane (RON)	91.0	93.8	95.7	97.5	99.2	100.7
Price (US$/gal)	$ 2.043	$ 2.043	$ 2.043	$ 2.043	$ 2.043	$ 2.043

Average CIF 2025 Prices

Source: Faro90

Thailand – Gasoline 95 – Increased Octane



Octane (RON)	95.0	97.9	99.7	101.2	102.8	104.2
Price (US$/gal)	$ 2.292	$ 2.292	$ 2.292	$ 2.292	$ 2.292	$ 2.292

Average CIF 2025 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**

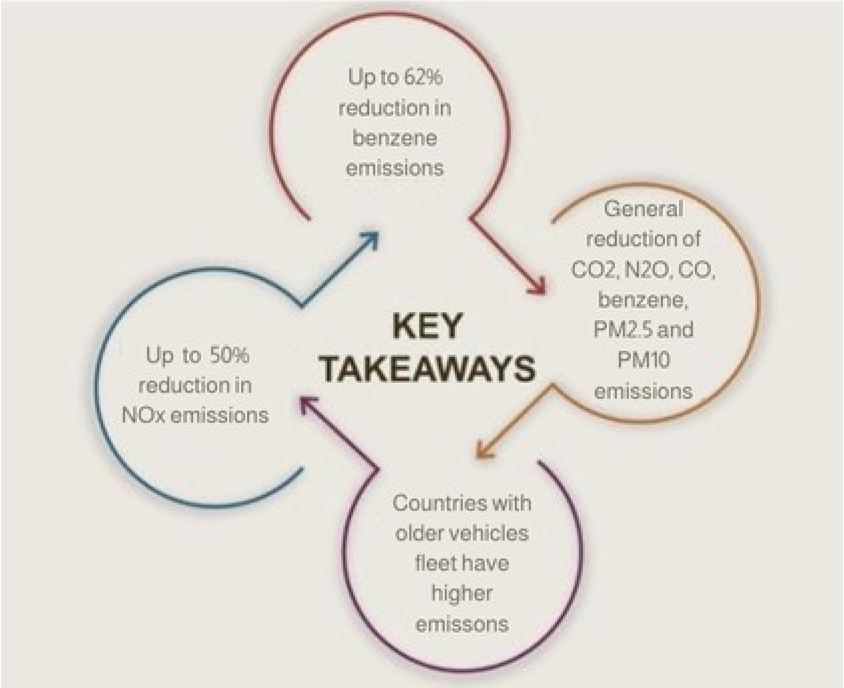
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



**Source: Bureau of transportation statistics.*

Gasoline Vehicle Fleet - Thailand

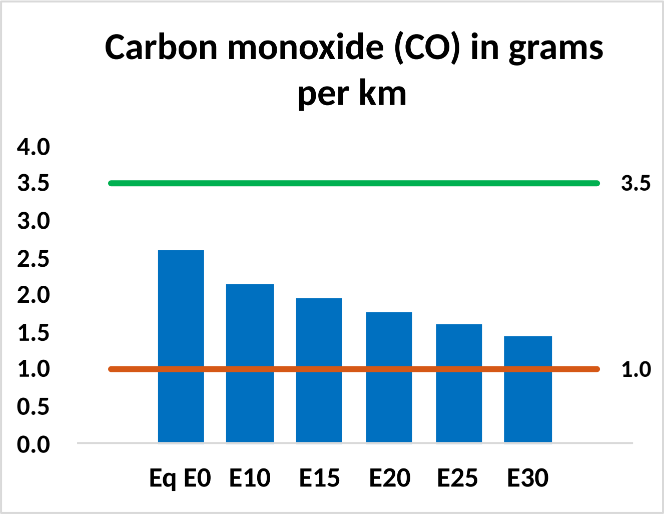
Type	Weight	Engine	Exhaust	Recovery	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	2,702
		Single FI	3-Ways		>8 years	77,373
		Multi FI	3-Ways		>8 years	338,735
			Euro III		>8 years	782,235
			Euro IV		>8 years	1,674,085
			Euro V		>8 years	1,320,165
			Euro 6		>8 years	2,813,146
					4-8 years	2,105,159
					<4 years	1,660,039
			Hybrid		All	416,913
		Electric			All	114,894
	Medium	Carburator	3-Ways	PCV Tank	>8 years	4,105
		Single FI	3-Ways		>8 years	97,929
		Multi FI	3-Ways		>8 years	539,680
			Euro III		>8 years	486,335
			Euro IV		>8 years	1,335,492
			Euro V		>8 years	546,791
			Euro 6		>8 years	989,322
					4-8 years	558,598
					<4 years	357,449
			Hybrid		All	190,225
		Electric			All	52,423
Sml Engines	Light	4-Cycle	None	PCV	>8 years	338,313
			Euro III		>8 years	1,179,819
			Euro IV		>8 years	1,699,542
			Euro V		>8 years	1,332,328
			Euro 6		>8 years	1,487,557
					4-8 years	1,092,581
					<4 years	1,463,627
			Hybrid		All	9,788
		Electric			All	56,423
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	2,938
		Fuel-Injection	3-Ways		>8 years	154,210
			Euro III		>8 years	32,474
			Euro IV		>8 years	255,191
			Euro V		>8 years	175,894
			Euro 6		>8 years	375,166
					4-8 years	174,338
					<4 years	97,167
			Hybrid		All	2
		Electric			All	2,961

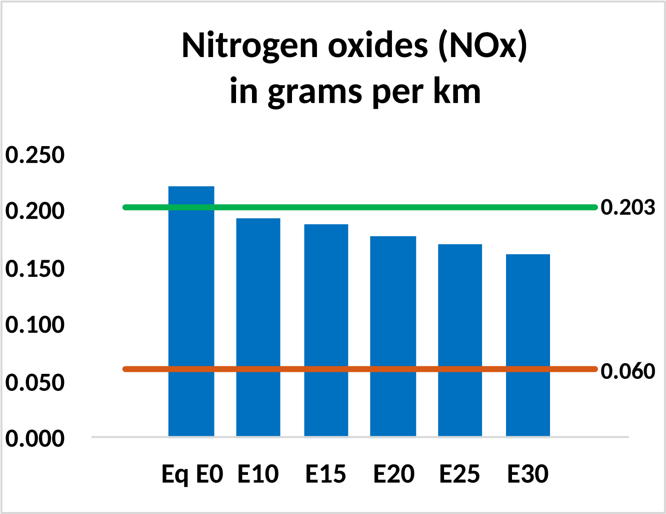
Vehicle Fleet: 26,394,115
Gasoline Vehicle Fleet: 23,593,936

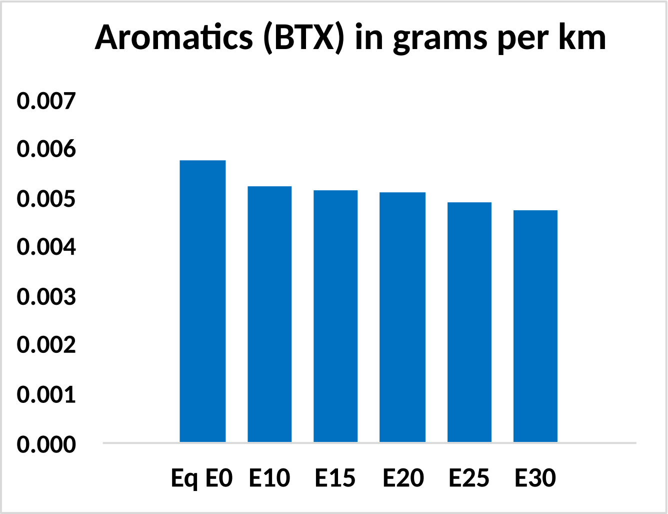
Source: datagov.mot.go.th

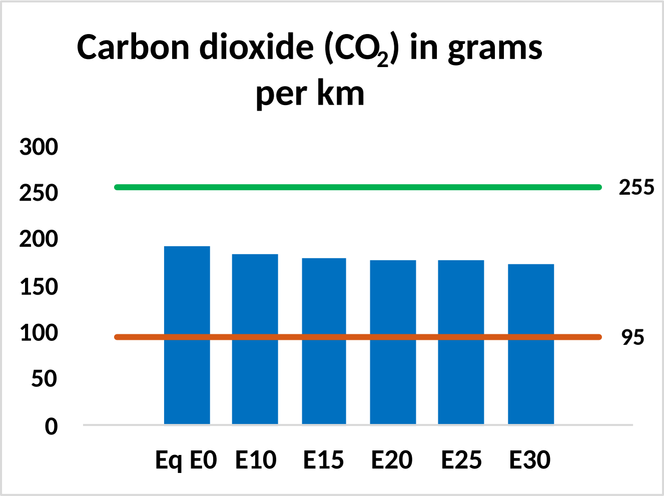
Thailand – Vehicular Emissions

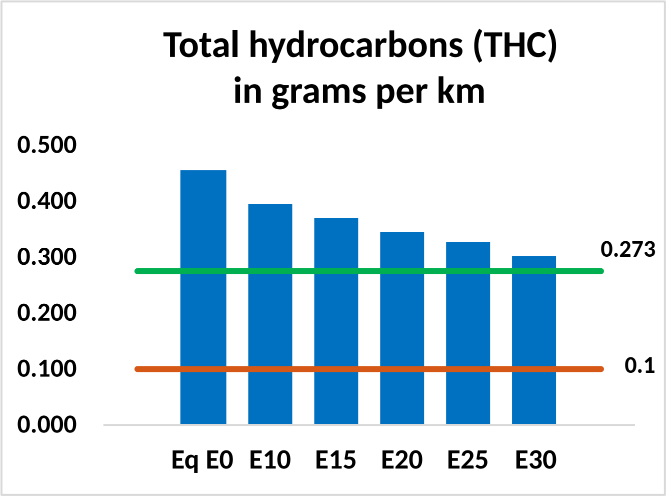
TIER III BIN 125 United States
Euro 6 Standard

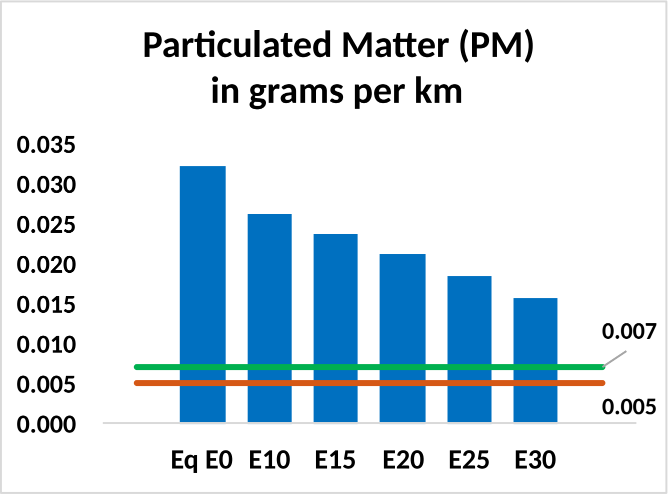












Source: Faro90

Thailand – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,272,173	1,161,670	1,051,167	958,136	863,375	790,165	712,435	-17%	-32%
CO2	94,740,194	92,304,017	90,003,184	88,193,646	87,298,216	86,891,621	85,289,959	-5%	-8%
VOC	223,716	208,572	193,427	181,417	169,515	160,419	148,085	-14%	-24%
NOx	109,552	101,153	95,401	92,476	87,758	84,063	79,710	-13%	-20%
SOx	1,084	1,002	920	851	784	712	637	-15%	-28%
NH3	34,761	34,417	34,073	34,234	34,620	34,890	35,115	-2%	0%
N2O	1,377	1,370	1,364	1,371	1,399	1,415	1,431	-1%	2%
CH4	52,376	52,376	52,376	53,161	54,313	55,309	56,251	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,358	1,289	1,221	1,172	1,126	1,092	1,043	-10%	-17%
Acetaldehyde	3,940	4,522	6,636	10,105	13,768	15,732	18,589	68%	249%
Formaldehyde	14,999	14,582	16,998	19,960	20,911	23,133	25,183	13%	39%
Benzene	2,835	2,724	2,581	2,540	2,519	2,409	2,332	-9%	-11%
PM2.5	5,761	5,237	4,713	4,252	3,802	3,318	2,812	-18%	-34%
PM10	10,064	9,149	8,234	7,429	6,641	5,797	4,913	-18%	-34%

Source: Faro90

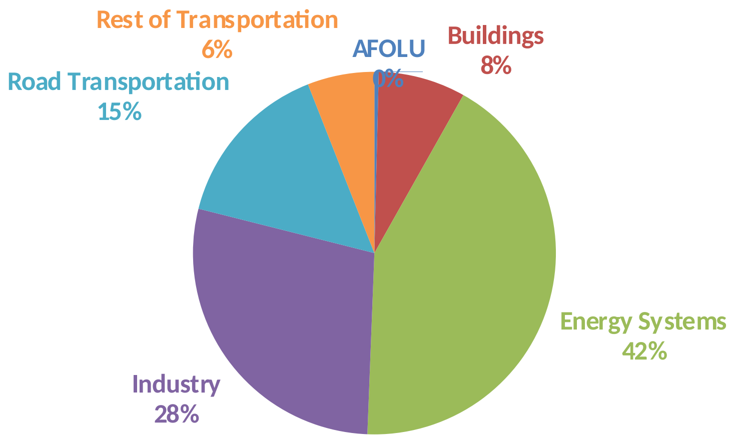
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

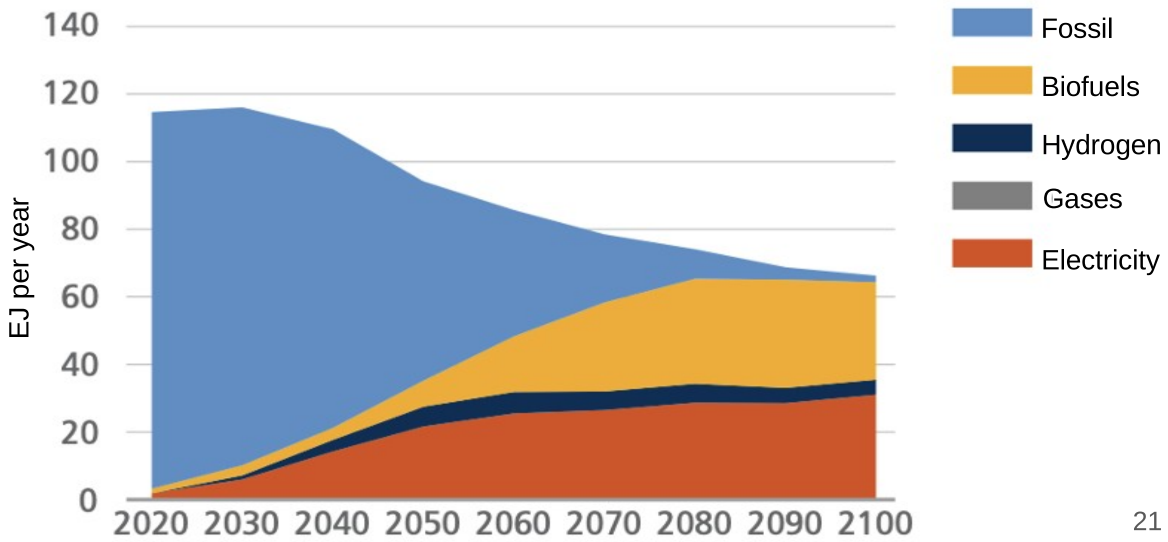
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

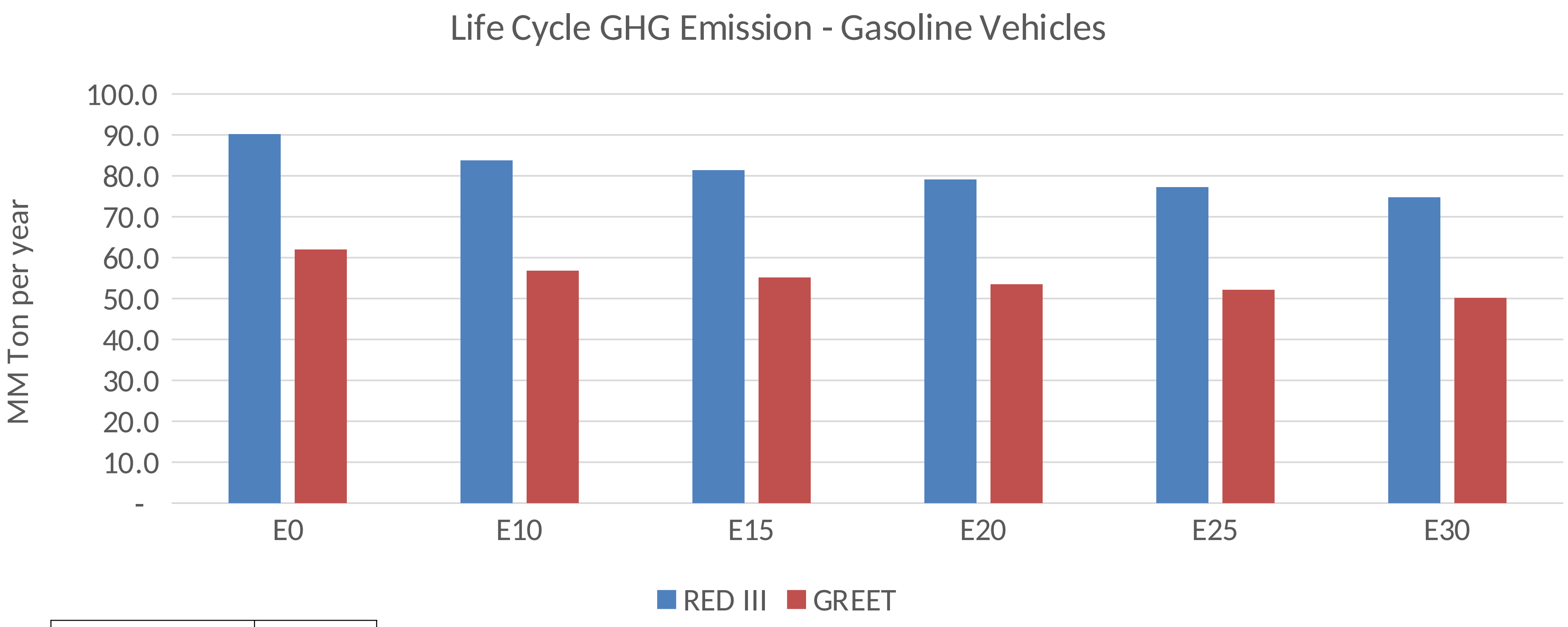
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Thailand – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	356.2
Target @ 2035 (MMT/year)	211.6
Delta	144.7

Source: Greet, RED II/III, climateactiontracker.org, F90